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Digital Image Processing

Assignment 03: Morphology

1. Objective

The objective of this experiment is to improve the quality of Global Thresholding segmentation results using Opening and Closing morphological operations. These methods aim to remove noise and fill small holes, thereby improving the overall shape and structure of the segmented objects. Additionally, this experiment aims to compare images obtained through segmentation alone with those processed using morphological operations to evaluate the impact of morphological post-processing on the clarity and structural integrity of the segmented objects.

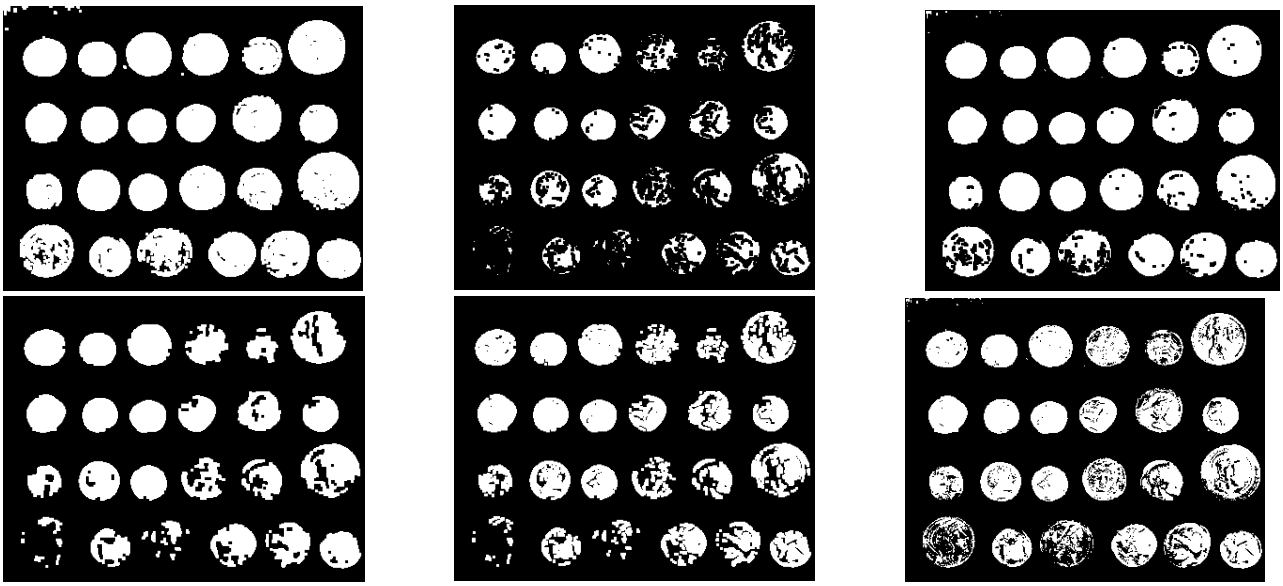
2. Segmentation

In this experiment, the input image consists of several coins against a black background containing noise. Since the input image is in grayscale, Global Thresholding was used as the segmentation method. A threshold value of 135 was applied, producing a binary image in which the coin objects are clearly distinguishable from the background. However, the segmentation result still contains noise in the background and several holes within the objects.

3. Morphology

To address these issues, morphological post-processing was applied to improve the quality of the segmentation results. Five morphological methods were implemented in this experiment: Dilation, Erosion, Opening, Closing, and a combination of Opening and Closing. Dilation is used to expand the foreground regions, while Erosion is used to reduce the size of objects. Furthermore, Opening aims to remove small foreground noise, whereas Closing is used to fill small holes or gaps within objects. The combination of Opening and Closing was applied to observe the effects of these two operations sequentially, allowing a comparison of object quality and structure before and after morphological post-processing.

4. Evaluasi



In order, the images above represent the results of morphological Dilation, Erosion, Closing, Opening-Closing combination, Opening, and the original segmentation without morphological processing. Based on the experimental results, the application of morphological post-processing produces different effects on the shape and structure of objects in the segmented image. The Dilation operation expands the foreground area, making the coins appear fuller and filling some gaps within the objects; however, background noise remains visible. In contrast, Erosion reduces the foreground area, causing several parts of the objects to shrink and lose important details. Furthermore, the Opening operation removes some small white noise but also eliminates certain parts of the coins due to the initial Erosion process. Meanwhile, Closing produces relatively better results in filling small holes while preserving the overall shape of the coins, although some larger holes remain visible. The combination of Opening and Closing does not necessarily produce more optimal results because object structures removed during Opening cannot be fully restored by the subsequent Closing operation. Overall, the experimental results demonstrate that the effectiveness of morphological post-processing depends on the characteristics of the input image, with Dilation producing the most visually distinct and complete object shapes in this experiment.